

CHAROTAR UNIVERSITY OF SCIENCE & TECHNOLOGY
1st Semester B. Tech. (EC/ME/CL)(CE/IT/EE)
University Examination – December 2010
CL102 Environmental Sciences

Date: 16/12/2010

Time: 10:30 AM To 01:30 PM

Maximum Marks: 70

Instructions:

1. The question paper comprises of two sections.
2. Attempt all questions.
3. Section I and II must be attempted in separate answer sheets.
4. Make suitable assumptions and draw neat figures wherever required.

SECTION – I

- Q - 1 (a)** Define the term ecology. Explain role of components of ecosystem. 03
- (b)** Explain energy transfer through an ecosystem with neat diagram. 04
- Q - 2 (a)** Write a short note on: Impact of Carbon cycle on environment. 03
- (b)** Write importance of food chain and food web in an ecosystem. 04
- (c)** Classify natural resources. 03
- (d)** Compare conventional and non- conventional energy resources. 04

OR

- Q - 2 (a)** Explain environmental significance of hydrological cycle and nitrogen cycle. 03
- (b)** Describe ecological pyramid of number. 04
- (c)** Write environmental impact due to overexploitation of water resources. 03
- (d)** Explain causes and effects of deforestation. 04
- Q - 3 Attempt any Two** 14
- (a)** Write factors affecting human settlement. Compare Malthusian theory and optimum population theory.
- (b)** Explain demographic transition theory.
- (c)** Define human settlement. Explain how overpopulation is acting as a pollutant.

(P.T.O)

SECTION - II

Q - 4 (a) Define the term Environment Engineering and write guiding principles of environmental studies. 03

(b) With the help of neat sketch explain lithosphere. 04

Q - 5 (a) Explain the anthropogenic sources of environmental pollution. 03

(b) What are the sources and effects of noise pollution? 04

(c) Explain the economic effects of air pollution on material. 03

(d) Describe various chemical characteristics of drinking water quality. Explain fluoride in detail. 04

OR

Q - 5 (a) Write short note on "Eutrophication". 03

(b) Explain the term water pollutant and describe organic water pollutants. 04

(c) What do you understand by ambient air quality standard? Describe Indian air quality standards. 03

(d) Explain the classification of air pollutants and describe according to its state. 04

Q - 6 Attempt any Two 14

(a) Explain the terms: Green House Effect & Acid rain.

(b) With the help of neat sketch explain the structure of atmosphere.

(c) Describe Turbidity and Hardness parameters of drinking water quality.
